

Business Report: Delinquency Risk Insights and Collections Strategy

Executive Summary

This report summarizes key findings from the analysis of Geldium's customer dataset and outlines recommendations to support a more targeted collections strategy. Using exploratory data analysis and a predictive modeling framework, several customer characteristics were identified as indicators of increased delinquency risk. These insights can help Geldium prioritize collection efforts, improve customer engagement, and reduce future delinquency rates through proactive intervention.

Summary of Predictive Insights

High-Risk Customer Segments

Customers displaying the following characteristics are more likely to become delinquent:

- Multiple missed payments
- High credit utilization
- High debt-to-income ratios
- Lower credit scores
- Elevated loan balances

These customers should be prioritized for early intervention because they show stronger indicators of financial stress and repayment difficulties.

Top 3 Risk Factors

1. **Missed Payments** – Previous missed payments are the strongest indicator of future delinquency.
 2. **Credit Utilization** – Customers using a high percentage of available credit are more likely to experience repayment challenges.
 3. **Debt-to-Income Ratio** – Customers with higher debt obligations relative to income face increased delinquency risk.
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Business Recommendation

SMART Recommendation

Implement a six-week targeted outreach program for customers identified as high risk based on missed payments, credit utilization, and debt-to-income ratio.

The program should use automated SMS reminders, email communications, and proactive collections outreach. The objective is to reduce 30-day delinquency rates by at least 10% within the next quarter.

Rationale

This recommendation is specific, measurable, actionable, relevant, and time-bound. It leverages existing communication channels, can be scaled efficiently, and focuses resources on customers most likely to benefit from early intervention. By engaging customers before delinquency worsens, Geldium can improve repayment outcomes while maintaining positive customer relationships.

Expected Business Impact

The proposed strategy is expected to:

- Improve collections efficiency by focusing efforts on higher-risk customers.
- Reduce delinquency rates through earlier intervention.
- Improve customer retention by providing support before accounts become severely delinquent.
- Enable more effective allocation of collections resources.

By prioritizing high-risk segments, Geldium can move from reactive collections management to a proactive, data-driven strategy.

Ethical and Responsible AI Considerations

Fairness Risk 1: Historical Bias

Historical customer data may contain patterns that unintentionally disadvantage certain groups.

Mitigation: Regularly review model performance across customer segments and conduct fairness audits to identify and address potential bias.

Fairness Risk 2: Proxy Bias

Variables such as location or other indirect indicators may unintentionally influence predictions in ways that create unfair outcomes.

Mitigation: Carefully review feature selection and monitor model outputs to ensure that predictions are based on financial risk indicators rather than demographic proxies.

Explainability

Model predictions should be explained in plain language. For example, a customer may be classified as higher risk because of multiple missed payments, high credit utilization, and elevated debt levels. Providing clear explanations helps stakeholders understand and trust AI-assisted decisions.

Responsible AI Use

AI predictions should support human decision-making rather than replace it. Collections decisions should remain transparent, regularly reviewed, and aligned with fairness and regulatory requirements.

Conclusion

The analysis identified missed payments, credit utilization, and debt-to-income ratio as key indicators of delinquency risk. A targeted outreach strategy focused on high-risk customers offers a practical opportunity to reduce delinquency while improving collections efficiency. By combining predictive analytics with responsible AI practices and human oversight, Geldium can make more informed, fair, and effective collections decisions.